



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

SmartSeg

The domain of the Project:
AI & ML

Team Mentor :
Mr.Gaurav Patel
Data Engineer- Celebal Technologies

Team Member:
Mr. Charchit Chauhan

Period of the project

April 2026 to June 2026



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Declaration

The project titled SmartSeg: Intelligent Customer Segmentation and Personalized Offer Recommendation System has been mentored by Mr. Gaurav Patel, organised by SURE Trust, from April 2026 to June 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:

Mr. Charchit Chauhan

Mr. Gaurav Patel
Data Engineer- Celebal Technologies

Mr. Gaurav Patel
Data Engineer- Celebal Technologies

Prof. Radhakumari
Executive Director & Founder
SURE Trust



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Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

SmartSeg is a comprehensive **Customer Segmentation and Personalized Marketing Platform** developed to assist retail businesses, particularly shopping malls, in understanding customer behavior and improving sales performance through intelligent data-driven strategies.

The system analyzes customer purchase data using **K-Means Clustering** to categorize customers into five meaningful segments: **VIP Customers, Loyal Customers, New Customers, At Risk, and Hibernating**. Based on these segments, it generates **personalized offers**, manages a **gamified loyalty program** with tier progression (Bronze, Silver, Gold), and provides rich interactive visualizations.

Built using modern technologies such as Python, Streamlit, Scikit-learn, Plotly, Groq AI, and MySQL (Railway Cloud), SmartSeg offers a user-friendly dashboard where mall owners can upload data, explore insights, run SQL queries, and interact with an AI assistant for business queries.

Key outcomes include improved customer retention, optimized marketing efforts, and actionable business intelligence — making it a practical and valuable tool for the retail industry.



Introduction

Background and Context The retail sector in India is highly competitive, with rapid growth in e-commerce and changing customer expectations. Malls and retail stores need advanced tools to understand customer preferences, predict buying behavior, and deliver personalized experiences. Traditional marketing approaches are becoming ineffective as customers demand more relevant offers.

Problem Statement Most mall owners still rely on generic promotions and lack deep insights into their customer base. This results in low customer retention, wasted marketing budgets, difficulty in identifying high-value customers, and poor personalization.

Goals of the Project The main goal was to develop an end-to-end intelligent system that can segment customers accurately, recommend personalized offers, manage loyalty rewards, and provide easy-to-understand business insights.

Scope

- Customer segmentation using Machine Learning
- Personalized offer recommendation
- Gamified loyalty program
- Interactive dashboard with filters and visualizations
- Groq AI chat assistant
- Support for CSV and MySQL database

Innovation SmartSeg stands out by integrating Machine Learning, Groq AI, gamified loyalty, and database support into a single, easy-to-use platform tailored for retail businesses.



Project Objectives

Clearly Defined Objectives

The primary goal of SmartSeg is to build a practical, intelligent system that helps mall owners understand their customers and improve business performance through data-driven personalization.

Specific Objectives:

1. To analyze customer purchase behavior and perform accurate customer segmentation using K-Means Clustering algorithm.
2. To develop a dynamic personalized offer recommendation engine that suggests relevant offers based on customer segments.
3. To implement a gamified loyalty program with a tier system (Starter, Bronze, Silver, Gold), loyalty points calculation, progress tracking, and rewards.
4. To create an interactive and visually rich dashboard using Streamlit that provides real-time business insights and trends.
5. To integrate Groq AI as an intelligent chat assistant, allowing mall owners to ask natural language questions about sales, customers, and segments.
6. To support flexible data input through CSV file upload and MySQL database connectivity for real-world usability.

Expected Outcomes and Deliverables

Expected Outcomes:

- Clear and meaningful customer segmentation with business interpretation.
- Improved customer retention through targeted personalized offers.
- Better decision-making for mall owners with actionable insights.
- A complete, ready-to-use tool suitable for real retail environments.



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Key Deliverables:

- Fully functional Streamlit web application with 7 interactive tabs.
 - K-Means clustering model with 5 customer segments.
 - Dynamic personalized offer recommendation system.
 - Gamified loyalty program with tier progression and visual progress bars.
 - Rich interactive visualizations and trend analysis.
 - Groq AI-powered chat assistant for business queries.
 - Support for both CSV upload and MySQL database.
 - Complete project documentation and source code on GitHub.
 - Live demonstration and presentation of the working system.
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- Complete project documentation and source code on GitHub.
 - Project demonstration and presentation.



Methodology and Results

The project was developed following a structured approach:

- **Data Understanding & Preparation:** Loading, cleaning, and exploring the Shopping Trends dataset.
- **Modeling:** Feature engineering and K-Means clustering for customer segmentation.
- **Business Logic Development:** Loyalty points calculation and personalized offer engine.
- **User Interface:** Interactive dashboard using Streamlit.
- **AI Integration:** Groq API for natural language queries.
- **Database:** MySQL for persistent data storage.

Tools and Software Used

- Programming Language: Python 3.13
- Web Framework: Streamlit
- Data Processing: Pandas, NumPy
- Machine Learning: Scikit-learn (K-Means, StandardScaler)
- Visualization: Plotly Express
- AI: Groq (Llama-3.3-70b-versatile model)
- Database: MySQL (Railway Cloud)
- Development Tools: VS Code, Git, GitHub

Data Collection Approach The project uses the publicly available **Shopping Trends Dataset** from Kaggle. It contains approximately 3,900 customer records with features like Age, Gender, Category, Purchase Amount, Season, and Review Rating. No primary data was collected.

Project Architecture The system is built with a modular layered architecture:

1. **Data Layer:** CSV upload or MySQL database
2. **Processing Layer:** Data preprocessing and K-Means clustering
3. **Business Logic Layer:** Loyalty system and personalized offers
4. **AI Layer:** Groq-powered chat assistant



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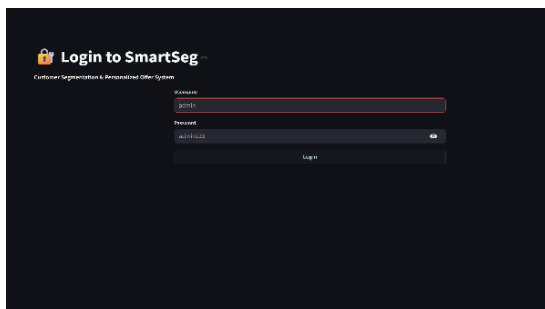
5. **Presentation Layer:** Interactive Streamlit dashboard

6. **Output Layer:** Visualizations, reports, and insights

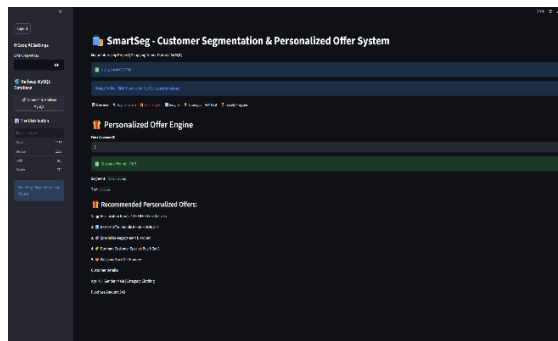
Results

- Successfully implemented 5 customer segments with clear business meaning.
- Dynamic personalized offers generated for each segment.
- Functional loyalty program with tier distribution.
- Interactive dashboard with filters and visualizations.
- Working Groq AI assistant for business queries.
- MySQL integration with Railway Cloud.

Final Project Working Screenshots

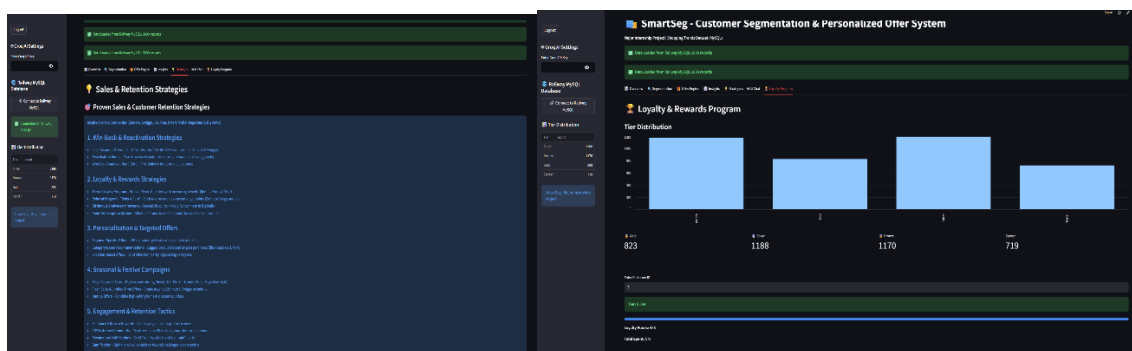
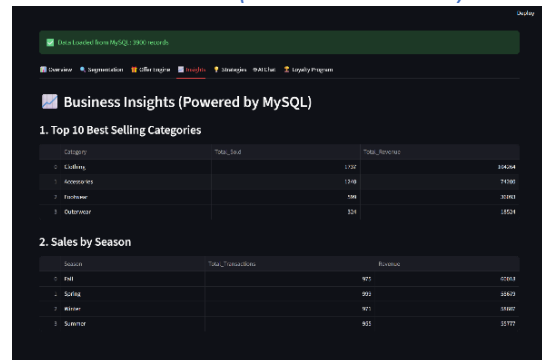
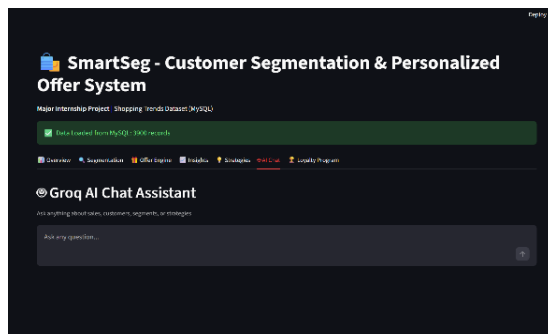


Customer ID	Age	Gender	Marital Status	Occupation	Income	Spouse Income	Children	Education	Household Income	Spouse Education	Spouse Income
1	35	Male	Married	Software Engineer	120000	80000	2	Graduate	200000	Graduate	100000
2	28	Female	Single	Marketing Executive	70000	0	0	Graduate	70000	Graduate	0
3	42	Male	Married	Business Owner	150000	100000	3	Graduate	250000	Graduate	120000
4	30	Female	Single	Teacher	60000	0	0	Graduate	60000	Graduate	0
5	38	Male	Married	Doctor	180000	120000	1	Graduate	300000	Graduate	150000
6	25	Female	Single	Freelance Writer	50000	0	0	Graduate	50000	Graduate	0
7	45	Male	Married	Retired	40000	30000	2	Graduate	70000	Graduate	30000
8	32	Female	Single	Consultant	90000	0	0	Graduate	90000	Graduate	0





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- Login Page
- Dataset Overview with multiple graphs
- K-Means Clustering Visualization
- Personalized Offer Engine
- Business Insights Dashboard
- Groq AI Chat Assistant
- Loyalty Program Interface

Project GitHub Link

GitHub Link: <https://github.com/charchit-chauhan/SmartSeg>

Streamlit App Link

<https://charchit-chauhan-smartseg-finalapp-xhgtmx.streamlit.app/>



Learning and Reflection

New Learnings (Technical)

- Machine Learning: Gained hands-on experience in unsupervised learning, specifically K-Means Clustering. Learned the importance of feature scaling using Standard Scaler and how to determine the optimal number of clusters.
- Data Analysis: Deepened knowledge of data preprocessing, feature engineering, and exploratory data analysis using Pandas and NumPy.
- Web Application Development: Learned to build interactive, professional dashboards using Streamlit— including session state management, caching, tabs, and file upload functionality.
- Data Visualization: Mastered Plotly Express to create insightful and interactive graphs such as scatter plots, box plots, histograms, and pie charts.
- AI Integration: Successfully integrated Groq AI (llama-3.3-70b-versatile model) to build a natural language chat assistant that can answer business questions based on the dataset.
- Database Management: Learned to connect Streamlit applications with MySQL (Railway) for real database operations.
- Version Control: Improved skills in Git and GitHub for project management and documentation.

New Learnings (Management & Soft Skills)

- Project planning, time management, and breaking down a complex project into manageable modules.
- Documentation skills — writing a professional project report with clear structure.
- Problem-solving under pressure, especially while debugging integration issues (MySQL, Groq API, Streamlit).
- Presentation and communication skills for explaining technical concepts to non-technical stakeholders.



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Overall Experience

Working on the SmartSeg project has been one of the most valuable learning experiences of my academic journey. It gave me the opportunity to apply classroom knowledge to a real-world problem in the retail sector.

The project challenged me to integrate multiple technologies — Machine Learning, Web Development, Database, and AI — into a single cohesive application. There were several debugging challenges, especially with MySQL connection and Groq AI integration, but overcoming them significantly boosted my confidence.

I am particularly proud of building a system that is not only technically sound but also practically useful for mall owners. The ability to upload datasets, view insights, get personalized offers, and chat with an AI assistant makes SmartSeg a complete business intelligence solution.

This project has strengthened my passion for data science and AI. It has prepared me for future roles in the industry by teaching me how to deliver end-to-end solutions that solve actual business problems.

Key Takeaway: Technology becomes truly powerful when it is applied to solve real-life problems and deliver value to users.



Conclusion and Future Scope

Conclusion

The SmartSeg project has successfully fulfilled its core objectives by developing a comprehensive, intelligent, and user-friendly Customer Segmentation and Personalized Offer Recommendation System. Using the Shopping Trends dataset, the system effectively analyzes customer behavior through key features such as Age, Purchase Amount, Previous Purchases, and Review Rating.

The implementation of K-Means Clustering enabled accurate segmentation of customers into five meaningful groups: VIP Customers, Loyal Customers, New Customers, At Risk, and Hibernating. The project delivers a modern Streamlit dashboard featuring interactive visualizations, a dynamic personalized offer engine, a gamified loyalty program with tier progression (Bronze, Silver, Gold), and a Groq AI-powered chat assistant for natural language business queries.

Key achievements include:

- Successful customer segmentation with clear business interpretation
- Personalized offer generation based on segments
- Interactive loyalty program with progress tracking and rewards
- Rich visualizations for trend analysis
- Flexible support for CSV upload and MySQL database connectivity
- AI assistant capable of answering domain-specific business questions

This project has demonstrated the practical application of Machine Learning, Data Analytics, and AI technologies to solve real-world retail challenges, providing mall owners with actionable insights for better decision-making, improved customer retention, and increased revenue.



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Future Scope of this Project

SmartSeg has strong potential for further development and real-world deployment. Future enhancements may include:

1. **Advanced Recommendation Engine:** Implement collaborative filtering or deep learning-based systems to recommend specific products to individual customers.
2. **Real-time Integration:** Connect with live POS/billing systems for real-time customer segmentation and instant personalized offers.
3. **Mobile Application:** Develop Android and iOS applications for customers to track loyalty points, redeem rewards, and receive push notifications.
4. **Predictive Analytics:** Add features such as customer churn prediction and Customer Lifetime Value (CLV) forecasting.
5. **Cloud Deployment:** Deploy the application on scalable cloud platforms like AWS, Heroku, or Google Cloud with proper authentication and security.
6. **Multi-language Support:** Extend support for regional Indian languages to increase accessibility.
7. **Enhanced Loyalty System:** Implement point redemption for vouchers, integration with payment gateways, and tier-based exclusive benefits.
8. **Advanced Security & Scalability:** Add role-based access control, data encryption, and support for large-scale retail chains.

With these improvements, SmartSeg can evolve into a full-fledged *Retail Business Intelligence Platform* suitable for medium to large shopping malls and retail chains across India.